

Are Small LLMs Ready for Coding Agents?

Can tight execution feedback turn bounded small-model actions into accepted workflows?

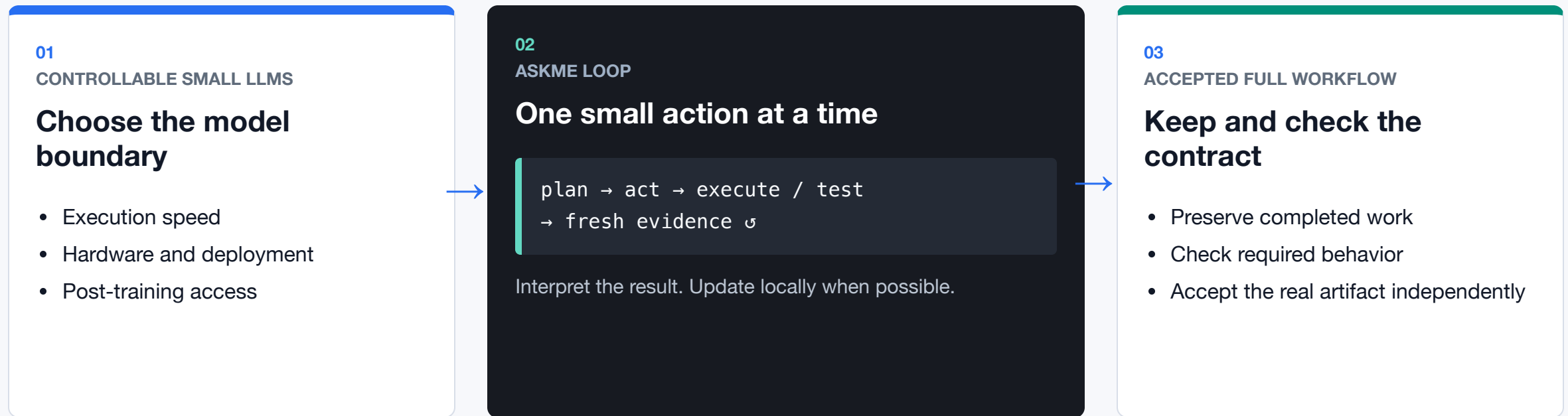
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A tight loop connects model control to full workflows

The model gets one bounded move; the harness keeps the end-to-end contract.



The bridge: small action surface + fresh execution feedback + independent workflow acceptance.

Assumes bounded actions, informative feedback, and independently testable success.

The code ran. The workflow still failed.

REQUIRED CONTRACT

```
write msg.h + main.c
build ./main
run ./main
observe REPLAN_OK
```

Source and header written

The requested program was correct.

/tmp/test compiled

A different output path was chosen.

/tmp/test ran successfully

It printed `REPLAN_OK`.

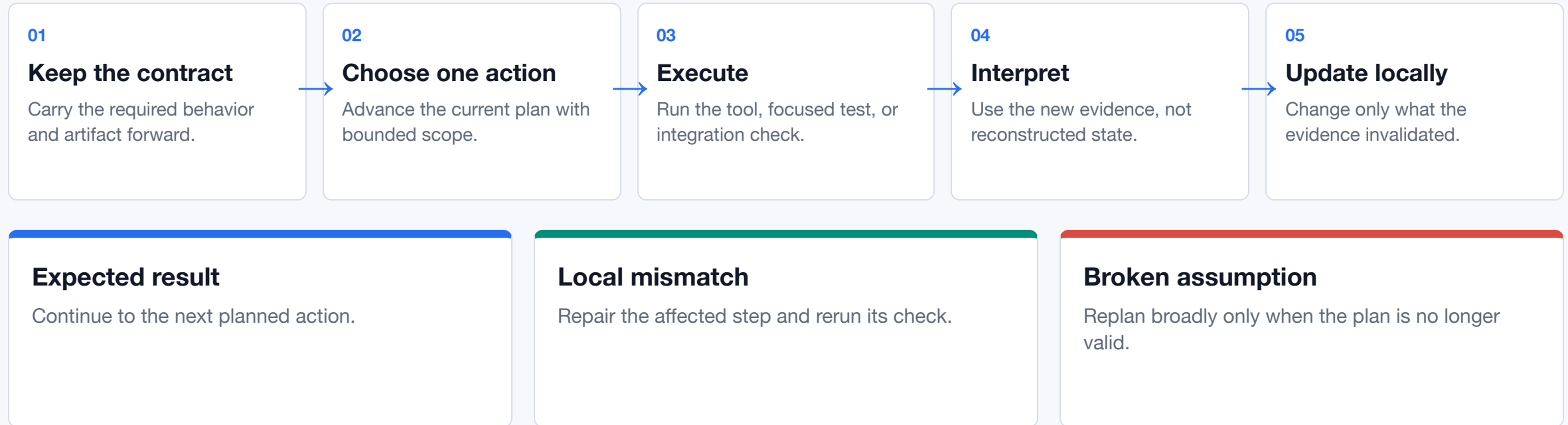
Agent reported completion

The independent acceptance test found no `./main`.

Successful actions + reported completion ≠ accepted artifact.

Current limit: acceptance scored this run after completion; the failure was not returned for recovery.

Hypothesis: update only what fresh evidence invalidates



Trajectory goal—not yet isolated: fewer repeated failures, fewer stuck steps, and less unnecessary plan churn.

Current result: 7 of 8 simple artifacts accepted

4 hosted model variants × 2 simple checks × 1 unseeded run/cell

8 / 8 agent complete

7 / 8 artifact accepted

Compatibility smoke · n=1/cell · descriptive only

MODEL	SHAPE	ARTIFACT BUILD	SCRIPT REPAIR
Gemma 4 26B A4B	♦ MoE 25.2B total · 3.8B active	ACCEPTED	ACCEPTED
Gemma 4 31B	• Dense 30.7B parameters	ACCEPTED	ACCEPTED
Qwen3.6-27B	• Dense 27B parameters	ACCEPTED	ACCEPTED
Qwen3.6-35B-A3B	♦ MoE 35B total · 3B active	NOT ACCEPTED wrong artifact path	ACCEPTED

Supported
AskMe ran all four variants; acceptance caught one false completion.

Not a clean model comparison
No pair isolates size: dense/MoE shape, active compute, run order, and trajectories changed. No Qwen-vs-Gemma, larger-vs-smaller, speed, reasoning, or reliability inference.

Test the reasoning policy—not model size

01

4 planned workflows

Semantic failures, visible feedback, held-out scoring; all must qualify.

02

1 frozen model route

Hold the model and harness conditions fixed.

03

Off vs gated × 3

Randomized repeats: $4 \times 2 \times 3 = 24$ total runs.

04

2 primary outcomes

Held-out acceptance and false completion across all valid runs.

Today: 1 / 4 workflows qualified · route and schedule pending · 0 measured runs

Scope: a large policy effect on these tasks—not Qwen vs Gemma, model size, architecture, or internal cognition.

Promising—with a tight harness.

Observed

8 / 8 reported completion. 7 / 8 simple artifacts were accepted. One completed run missed its deliverable.

Supported

Harness compatibility across four hosted variants; execution evidence plus independent acceptance is a credible direction.

Still open

Feature/app readiness, reasoning impact, Qwen vs Gemma, model-size effects, reliability, and local performance.

Make the interface easier to use.
Keep success tied to real behavior.

The smoke exercises the measurement path; readiness needs repeated held-out workflows.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data